Chem-E Car Propulsion Mechanism: Thermal Electric Generators

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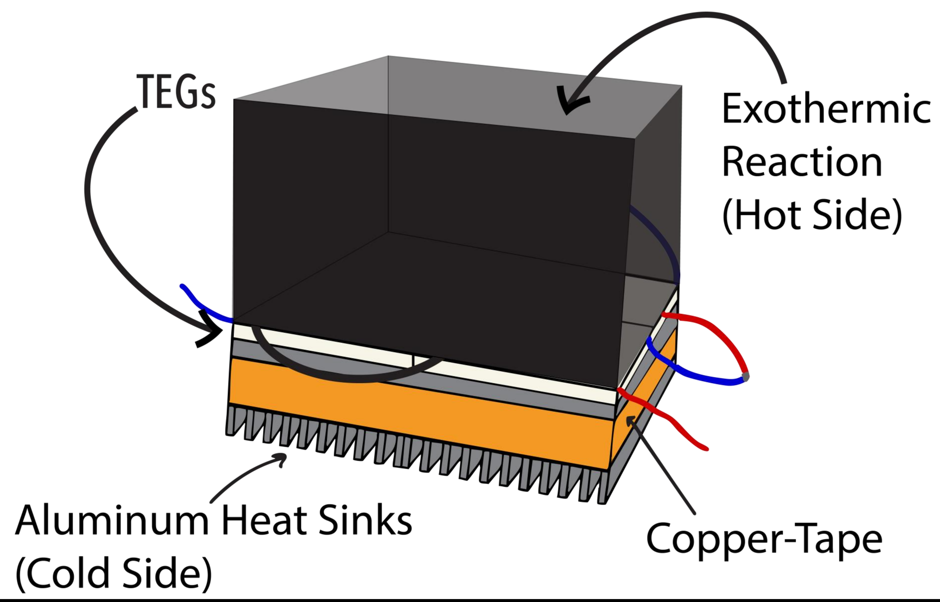
Abstract: The Chem-E-Car Competition tasked students with building a vehicle powered and controlled by chemical reactions. Our team developed a propulsion system using thermoelectric generators (TEGs), which convert heat from an acid-base reaction and cooling from dry ice into electrical energy. We analyzed performance by testing different TEG types, quantities, and chemical reactions, ultimately designing a 3D-printed carbon fiber container with two hot boxes and integrated heat sinks. This innovative setup contributed to CSULB's successful performance at the Western Regional Conference.

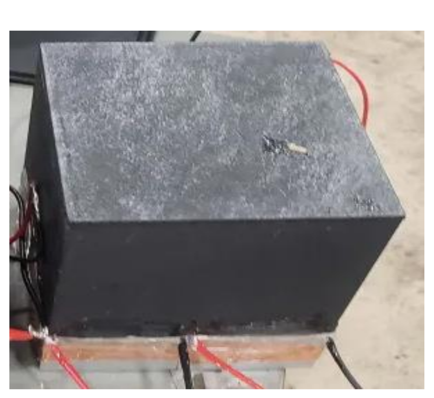


Apparatus and Procedure:

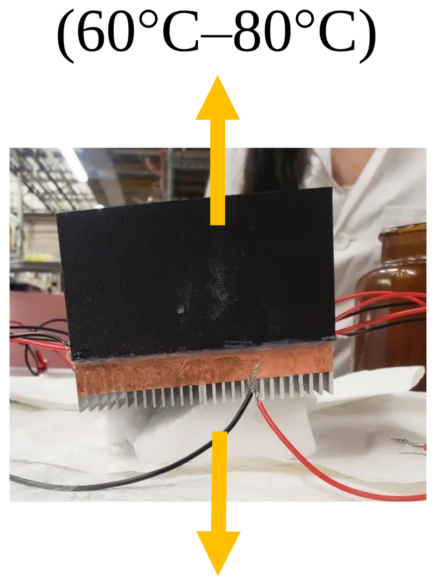
1. Acid-Base reaction preparation for exothermic reaction.
   a. Always wear full PPE: gloves, acid-resistant apron, googles, and face shield.
   b. Prepare the 3M solution of HCl.
   c. Add equimolar NaOH to 3M HCl to each container to complete exothermic reaction.

$$HCl_{(aq)} + NaOH_{(s)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} - 56.57\ kJ/mol$$

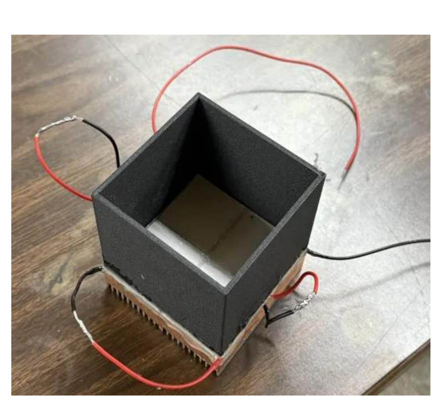
   d. Turn the switch on to let the pump mix the HCl with the NaOH and have constant stirring in the containers.
2. TEG Wiring and Testing
   a. Connect four TEGs in series per hot box to add voltage output.
   b. Use insulated wiring and check all connections before powering the system.
3. System Design and Assembly
   a. Use 3D-printed carbon fiber containers: hot box for the reaction and cold box for dry ice.
   b. Epoxy used to seal TEGs to the hot box; thermal paste applied evenly to attach heat sinks
   c. A pump was added to circulate the reaction, and a manual switch was installed for control.



Acid/Base (60°C–80°C)



Dry Ice (-78°C)



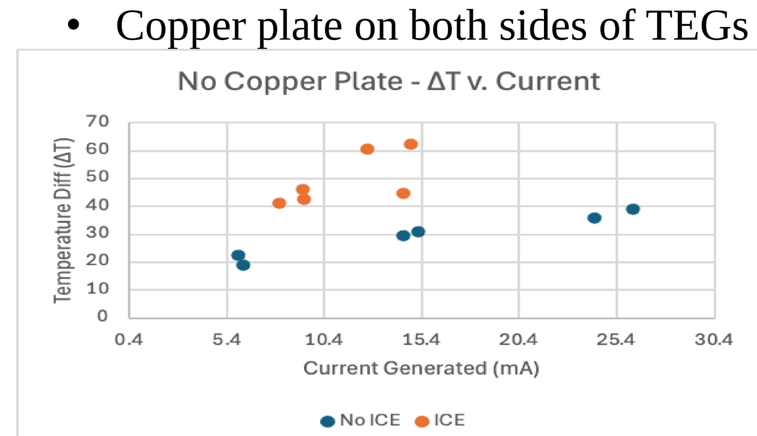
Safety Concerns :

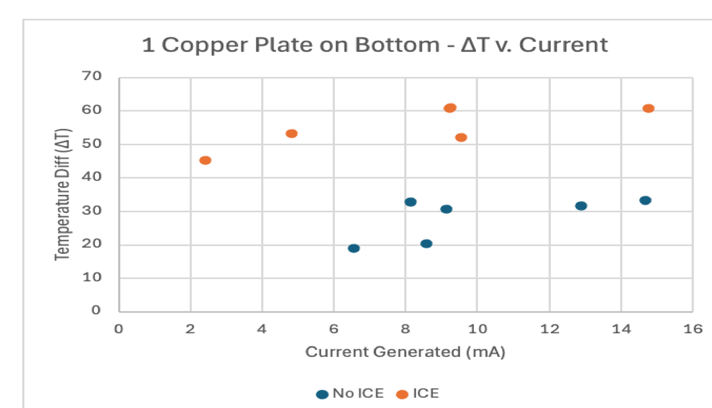
- Wear gloves, goggles, and lab coats at all times.
- Conduct reactions (e.g., acid dilution) in a fume hood.
- Neutralize and dispose of chemical waste per hazardous waste protocols.
- Insulate wires and secure all electrical connections.
- Report spills or accidents immediately.

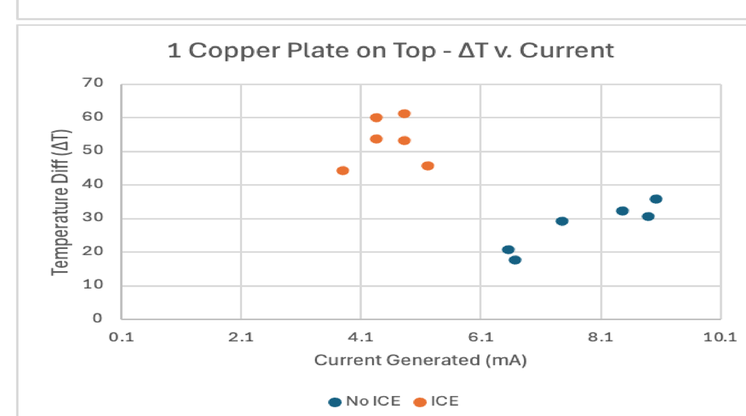
Discussion:

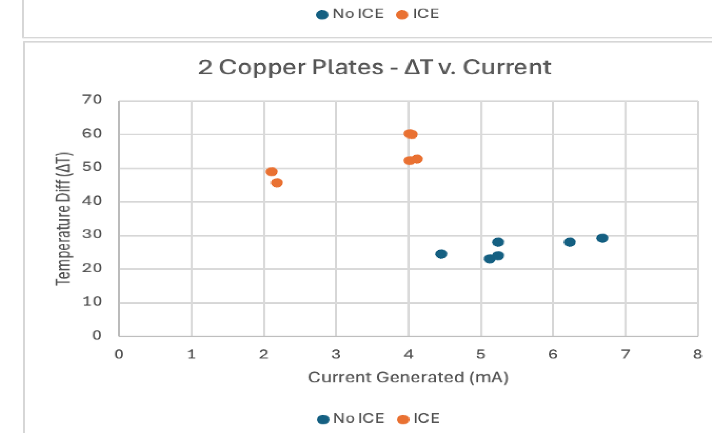
Early Research

- We used hot and ice water as substitutes for hot side and cold side.
- Design: 3D printed box with heat sinks and TEGs
- We tested voltage and current on 4 TEG sandwich setups with:
  - No copper plates
  - Copper Plate on top of TEGs
  - Copper plate on bottom of TEGs
  - Copper plate on both sides of TEGs









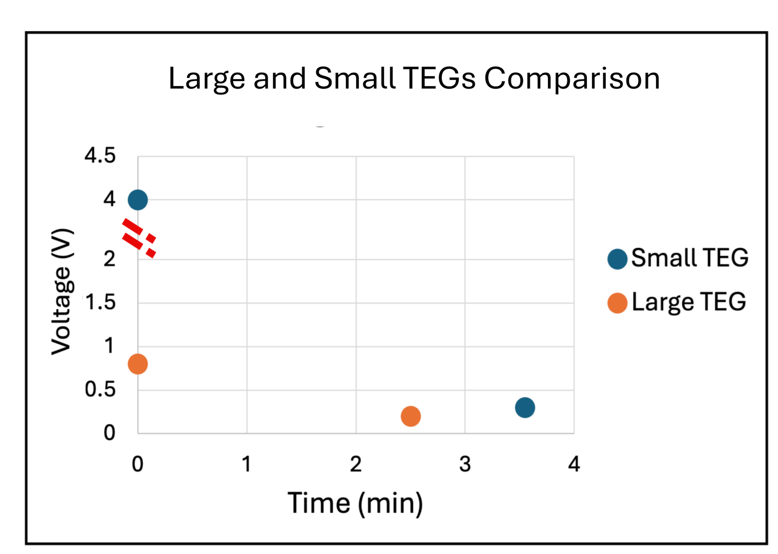
Early Key Findings

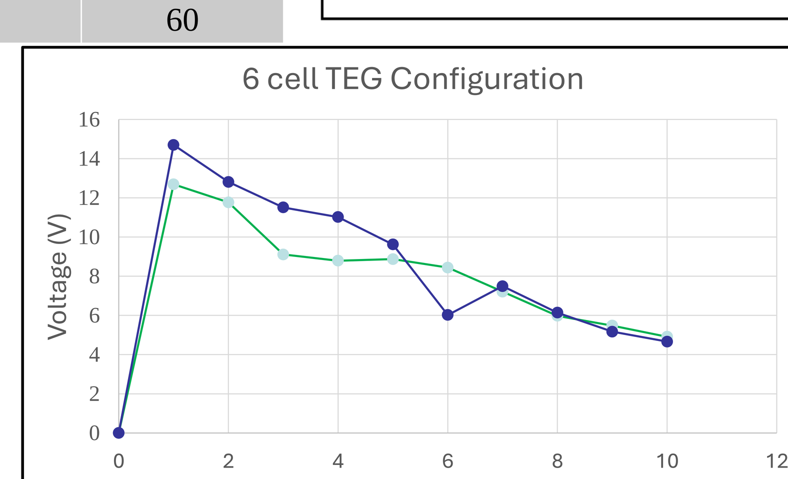
- Using no copper plates provided the best voltages due to less contact resistance.
- Carbon fiber filament enclosure proved durable to strong acid.

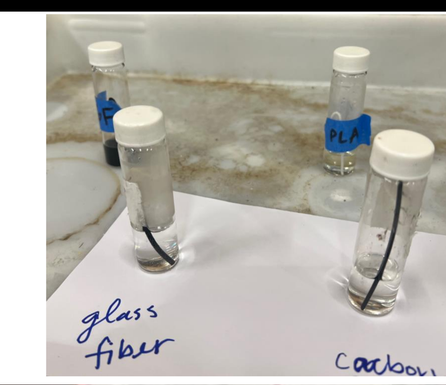
Later Research (Dry Ice, Pumps, Direct Contact)

- We finalized our design by applying acid/base reaction directly on TEG.
- We tested :
  - Quicklime as Optional Exothermic Reaction
  - Large vs Small TEGs
  - 6 Cell TEG Configuration

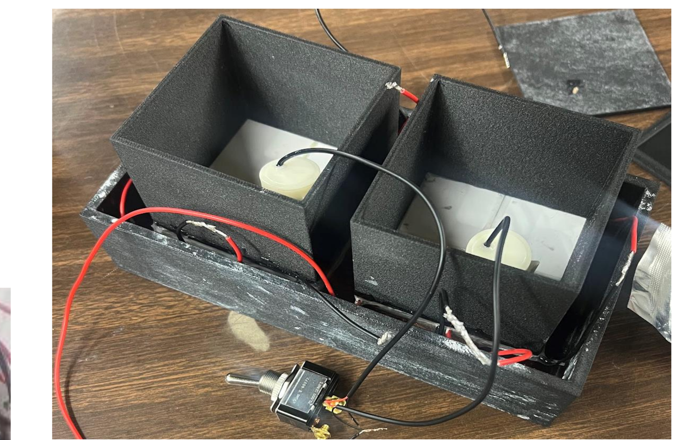
TEG - Model	SP1848-27145	TGMT-19W-4V
Open Circuit Voltage (V)	4.8	8.4
Operating Temperature (°C)	0 to 150	0 to 400
Maximum Temperature (°C)	150	300
Length (mm)	40	56
Width (mm)	40	56
Height (mm)	3.6	5
Weight (g)	30	60







FINAL DESIGN



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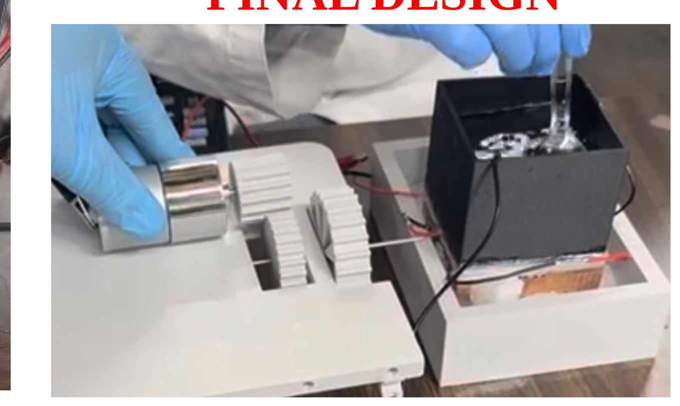

Last Key Findings

- Small TEGs outperformed large TEGs.
- Optimal performance achieved with 8 TEGs and 250g–350g of dry ice.
- Acid-base reactions are more stable lighter compared to CaO.
- Copper plates reduced performance and should be avoided.

System Behavior

The graph shows voltage and current output over 10 minutes. Initial values were 24 V and 1.3 A, declining to 3 V and 0.204 A. This trend reflects decreasing temperature difference and system efficiency over time, highlighting that the design is best suited for short bursts of power rather than sustained operation.

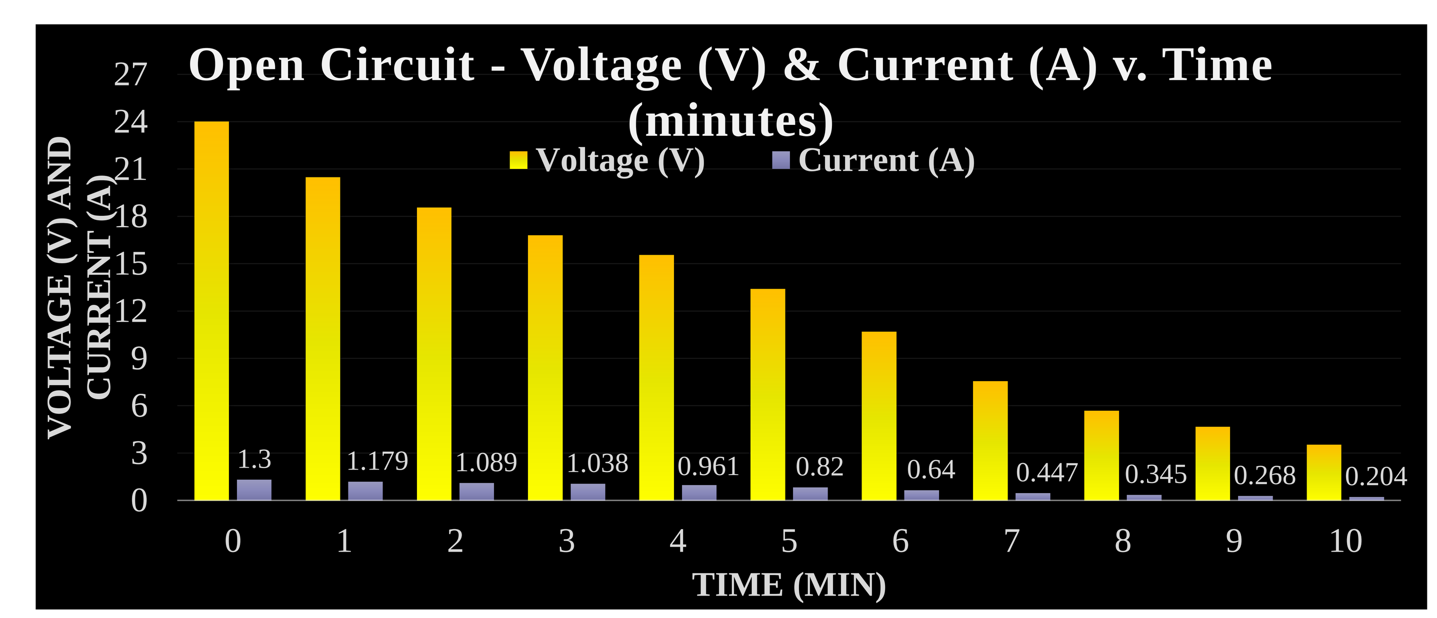

Figure 1: Voltage (V) and Current (A) vs. Time (min) of ThermoElectric Battery

Conclusion:

Over the semester, we successfully developed a thermoelectric propulsion system powered by an acid-base reaction to drive a small-scale vehicle that utilizes a 12V DC motor. The optimal design featured 8 TEGs (4 per hot box), a 3D-printed carbon fiber assembly, NaOH and 3M HCl as the heat source, and 250g –350g of dry ice as the cold sink. This design proved effective for short-duration applications and contributed to CSULB's success at the Western Regional Conference where the car ran a distance of 17.23 meters in 2 minutes.

Future Research:

- As students move onto the Nationals, they should extend voltage and current by either adjusting molarity of the acid or use better insulator than carbon fiber.
- The car should incorporate a voltage sensor (HiLetgo Voltage Detection) with OLED (I2C IIC) to view Voltage and Current generated by the car.
- Add a capacitor to the car to store charge from propulsion device and discharge it as a specified voltage (12V) for increased speed accuracy.
- Consider power calculations over 30 m distance to determine car performance.

Acknowledgement:

-Dr. Kang, Dr. Mendez, Mr. Minh, and Marathon Petroleum.

-*Fundamental of Heat and Mass Transfer*, Bergman, Lavine, 8th Ed.